

A conditional Hilbert-factorization reduction for the biased-coin Tylli problem

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Status

This is a conditional reduction for the final open problem in Johnson–Schechtman [1]. It does not claim a full solution. It proves that the biased-coin operator T_ε satisfies the Tylli and Weak Tylli conclusions once one proves a concrete uniform resolvent multiplier estimate for one smaller biased-coin operator T_δ .

Source problem

Johnson and Schechtman define T_ε on $L_p(\Delta)$, where $\Delta = \{-1, 1\}^\mathbb{N}$ is the Cantor group with Haar measure, as convolution by the ε -biased coin. Equivalently, if w_A denotes the Walsh character indexed by a finite set $A \subset \mathbb{N}$, then

$$T_\varepsilon w_A = \varepsilon^{|A|} w_A.$$

At the end of Section 5 they ask the following.

L_p , which implies that X_0 is also complemented in L_p .

We conclude this section with the open problem that started us thinking about ℓ_p -strictly singular operators.

Problem 1 *Let $1 < p < 2$ and $0 < \varepsilon < 1$. On $L_p(\Delta)$, does T_ε satisfy the conclusion of the Tylli Conjecture or the Weak Tylli Conjecture?*

Of course, the answer to Problem 1 is “yes” when $\varepsilon = \varepsilon(p)$ is sufficiently small, since then T_ε maps L_p into L_2 .

Thus, for $1 < p < 2$ and $0 < \varepsilon < 1$, the problem is whether T_ε lies in the norm closure of the operators factoring through Hilbert space, or at least satisfies the corresponding Weak Tylli conclusion after composition with an isometric embedding into ℓ_∞ . The source paper notes that the answer is already yes when ε is sufficiently small, because then $T_\varepsilon : L_p(\Delta) \rightarrow L_2(\Delta)$ is bounded [1].

Conditional dependency

Fix $1 < p < 2$ and $0 < \delta < 1$. Let Q_t be the Walsh multiplier

$$Q_t w_A = \frac{t}{t + \delta^{|A|}} w_A, \quad t > 0.$$

Conditional Dependency 1 (positive-resolvent Walsh multiplier bound). *There is a constant $C = C(p, \delta)$ such that*

$$\|Q_t\|_{L_p(\Delta) \rightarrow L_p(\Delta)} \leq C$$

for every $t > 0$.

Equivalently, $t(tI + T_\delta)^{-1}$ is uniformly bounded on $L_p(\Delta)$. This is implied by sectoriality of T_δ of angle $< \pi$, but the packet records the smaller estimate actually used in the proof.

Idea of the proof

Choose $\delta < \varepsilon$ so small that $T_\delta : L_p \rightarrow L_2$ is bounded. Then T_δ factors through Hilbert space. Put

$$\alpha = \frac{\log \varepsilon}{\log \delta} \in (0, 1),$$

so that $\delta^\alpha = \varepsilon$. The multiplier bound lets us form the Balakrishnan root integral

$$B_\alpha = \frac{\sin(\pi\alpha)}{\pi} \int_0^\infty t^{\alpha-1} T_\delta(tI + T_\delta)^{-1} dt$$

in operator norm. Each integrand still factors through Hilbert space, because it is T_δ followed by a bounded operator. On each Walsh character the scalar integral gives $(\delta^{|A|})^\alpha = \varepsilon^{|A|}$, so B_α is exactly T_ε . Thus the only missing ingredient is the uniform resolvent multiplier estimate.

Lemma 1 (Balakrishnan multiplier under Assumption 1). *Assume 1. For every $0 < \alpha < 1$, the operator*

$$B_\alpha = \frac{\sin(\pi\alpha)}{\pi} \int_0^\infty t^{\alpha-1} T_\delta(tI + T_\delta)^{-1} dt$$

is a well-defined bounded operator on $L_p(\Delta)$, with convergence in operator norm, and

$$B_\alpha w_A = (\delta^{|A|})^\alpha w_A$$

for every Walsh character w_A .

Proof. Assumption 1 gives boundedness of $Q_t = t(tI + T_\delta)^{-1}$. Hence

$$T_\delta(tI + T_\delta)^{-1} = I - Q_t$$

is bounded uniformly for $0 < t \leq 1$. This gives integrability near 0, because $t^{\alpha-1}$ is integrable there. For $t \geq 1$,

$$T_\delta(tI + T_\delta)^{-1} = t^{-1} T_\delta Q_t,$$

so its norm is at most $t^{-1} \|T_\delta\| C$, and $t^{\alpha-2}$ is integrable at infinity. Therefore the Bochner integral converges in operator norm.

On the one-dimensional span of a Walsh character w_A , the integral reduces to the classical scalar identity

$$\frac{\sin(\pi\alpha)}{\pi} \int_0^\infty t^{\alpha-1} \frac{\lambda}{t + \lambda} dt = \lambda^\alpha, \quad \lambda > 0, \quad 0 < \alpha < 1,$$

with $\lambda = \delta^{|A|}$. This proves the asserted action on Walsh characters. $\square$

Theorem 1 (conditional Hilbert-factorization conclusion). *Let $1 < p < 2$ and $0 < \delta < \varepsilon < 1$. Assume that $T_\delta : L_p(\Delta) \rightarrow L_2(\Delta)$ is bounded and that Assumption 1 holds for this δ . Then*

$$T_\varepsilon \in \overline{\Gamma_2}(L_p(\Delta)),$$

the norm closure of the operators on $L_p(\Delta)$ that factor through Hilbert space. Consequently T_ε satisfies the Tylli conclusion and the Weak Tylli conclusion.

Proof. Since $T_\delta : L_p \rightarrow L_2$ is bounded and the inclusion $L_2(\Delta) \hookrightarrow L_p(\Delta)$ is bounded on a probability space when $p < 2$, T_δ factors through the Hilbert space $L_2(\Delta)$. Let $S = T_\delta$.

Set $\alpha = \log \varepsilon / \log \delta$. Because $0 < \delta < \varepsilon < 1$, we have $0 < \alpha < 1$, and $\delta^\alpha = \varepsilon$. By Lemma 1,

$$B_\alpha = \frac{\sin(\pi\alpha)}{\pi} \int_0^\infty t^{\alpha-1} S(tI + S)^{-1} dt$$

is bounded on $L_p(\Delta)$ and satisfies

$$B_\alpha w_A = (\delta^{|A|})^\alpha w_A = \varepsilon^{|A|} w_A = T_\varepsilon w_A.$$

Walsh polynomials are dense in $L_p(\Delta)$, and both B_α and T_ε are bounded, so $B_\alpha = T_\varepsilon$.

It remains to check that B_α belongs to the closed Hilbert-factor ideal. For each $t > 0$, the operator

$$S(tI + S)^{-1} = SQ_t/t$$

factors through Hilbert space because S does and Q_t/t is a bounded operator on L_p . The integral defining B_α is the norm limit of Riemann sums of such Hilbert-factor operators. Since $\overline{\Gamma_2}(L_p)$ is a closed operator ideal, $B_\alpha \in \overline{\Gamma_2}(L_p)$.

This proves the Tylli conclusion. If $J : L_p(\Delta) \rightarrow \ell_\infty$ is an isometric embedding, then JT_ε is the norm limit of the compositions of J with Hilbert-factor operators. Thus JT_ε lies in the closure of the operators from $L_p(\Delta)$ to ℓ_∞ that factor through Hilbert space, which is the Weak Tylli conclusion. $\square$

Corollary 1 (what remains for a full solution). *For fixed $1 < p < 2$, if Assumption 1 holds for some $\delta > 0$ for which $T_\delta : L_p(\Delta) \rightarrow L_2(\Delta)$ is bounded, then every T_ε , $\delta < \varepsilon < 1$, has an affirmative answer to the Johnson–Schechtman problem.*

Verification and novelty notes

The proof is purely analytic after Assumption 1. The verification focus is therefore the multiplier family $m_t(n) = t/(t + \delta^n)$. A bounded sectorial functional calculus for T_δ would imply the assumption, but standard Ritt/Stolz calculi for sampled semigroups do not by themselves give the needed root z^α , because z^α is not holomorphic at 0 when $0 < \alpha < 1$.

Cheap run-index searches found no previous packet for arXiv:0708.0560. The nearby Johnson–Schechtman closed-ideal packets concern other strictly singular operators and do not answer this biased-coin problem. A bounded local and web search for the exact biased-coin/Tylli phrases found the source problem and relevant functional-calculus background, but no later explicit answer to this question.

References

References

- [1] W. B. Johnson and G. Schechtman, *Multiplication operators on $L(L_p)$ and ℓ_p -strictly singular operators*, arXiv:0708.0560.
- [2] C. Le Merdy, *H^∞ functional calculus and square function estimates for Ritt operators*, arXiv:1202.0768.
- [3] F. L. Schwenninger, *Functional calculus estimates for Tadmor–Ritt operators*, arXiv:1507.00049.